

Log Monitoring & Analysis

Objective:

The objective of this task is to monitor and analyze system logs to identify suspicious activity, detect anomalies, correlate security events, and document findings using security monitoring principles similar to those used in a Security Operations Center (SOC).

Tools:

Windows XP Firewall Logs (Windows Logs)
Linux-based Attacker System (Kali Linux)
Virtual Lab Network
SIEM Concept Reference: Splunk (theoretical)

Lab Environment:

The analysis was conducted in a controlled virtual lab environment consisting of:

Attacker Machine: Kali Linux - 192.168.136.128

Target System: Windows XP - 192.168.136.129

Network Range: 192.168.136.0/24

Windows Firewall logging was enabled on the Windows XP system with the following options:

- Log dropped packets
- Log successful connections

These logs are commonly collected and analyzed in SIEM platforms such as Splunk for security monitoring and incident detection.

Log Type and Structure:

The Windows Firewall log records network-level security events including:

- Date and time
- Action (DROP, OPEN, CLOSE)
- Protocol (TCP, UDP)
- Source and destination IP addresses
- Source and destination ports

- TCP flags

These logs provide visibility into connection attempts, blocked traffic, and allowed communications, making them useful for detecting reconnaissance and pre-authentication attack activity.

Log Analysis and Findings:

1. Background Network Traffic (NetBIOS – UDP 137):

The logs contain repeated dropped UDP packets targeting port 137:

DROP UDP 192.168.136.1 → 192.168.136.255 137

Analysis:

Port 137 is associated with NetBIOS Name Service.

The traffic represents broadcast activity common in Windows networks.

The firewall correctly dropped unsolicited broadcast packets.

Assessment:

Normal background traffic with no malicious intent detected.

2. Suspicious FTP Connection Attempts (TCP 21):

Multiple blocked TCP connection attempts targeting the FTP service were observed:

DROP TCP 192.168.136.128 → 192.168.136.129 46692 → 21 S

DROP TCP 192.168.136.128 → 192.168.136.129 46694 → 21 S

Analysis:

The source IP `192.168.136.128` attempted to initiate multiple connections to port 21.

TCP SYN flags indicate connection initiation attempts.

Repeated attempts suggest reconnaissance or service discovery behavior.

Assessment:

Suspicious activity consistent with port scanning and pre-authentication probing.

3. Telnet Scan Attempts (TCP 23):

Additional blocked attempts targeted the Telnet service:

DROP TCP 192.168.136.128 → 192.168.136.129 64097 → 23 S

DROP TCP 192.168.136.128 → 192.168.136.129 64099 → 23 S

Analysis:

Telnet is a legacy service that uses authentication and is often targeted for brute-force attacks.

Multiple SYN packets from the same source indicate probing behavior.

The firewall successfully prevented access.

Assessment:

Unauthorized access attempts targeting an insecure authentication service.

4. Allowed and Closed Connections (RDP – TCP 3389):

The log also recorded legitimate allowed connections:

OPEN-INBOUND TCP ... → 3389

CLOSE TCP ... → 3389

Analysis:

Port 3389 is used by Remote Desktop Protocol (RDP).

The firewall allowed and logged the connection correctly.

Proper session closure indicates normal system behavior.

Assessment:

Legitimate authenticated activity.

Authentication and Failed Login Relevance:

While no successful or failed login attempts were directly recorded in the firewall log, the blocked FTP and Telnet connection attempts represent pre-authentication attack activity.

Such reconnaissance commonly precedes brute-force or credential-based attacks and is an important indicator in log monitoring.

Event Correlation:

Correlation of events revealed:

A single source IP (192.168.136.128)

Multiple targeted services (FTP and Telnet)

Repeated attempts within a short time window

This pattern indicates coordinated reconnaissance activity rather than random traffic.

SIEM Perspective and Alert Example:

If these logs were ingested into a SIEM such as Splunk, the following alert could be generated:

Alert Name: Suspicious Port Scanning Activity

Description: Multiple blocked TCP SYN packets from a single source IP targeting multiple services.

Severity: Medium

MITRE ATT&CK Mapping: Reconnaissance – T1046 (Network Service Discovery)

Recommended Action: Monitor or block the source IP if activity persists.

Recommendations:

- Continue monitoring firewall and system logs.
- Disable legacy services such as Telnet if not required.
- Forward logs to a centralized SIEM for real-time alerting and correlation.
- Investigate repeated reconnaissance attempts as early indicators of attack.